

CoRoT-2b

R-0183

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240710 · created 2026-07-16T19:32:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460501.8717 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.024
Transit depth (Rp/Rs)^2	2.2e-06 +/- 0.00071 [%]
Orbital Inclination (inc)	83.21 +/- 2.06
Airmass coefficient 1 (a1)	1.417 +/- 0.04
Airmass coefficient 2 (a2)	-0.275 +/- 0.07
Scatter in the residuals of the lightcurve fit is	2.89 %
Best Comparison Star	#3 - [261, 386]
Optimal Method	PSF photometry
Transit Duration (day)	0.046 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.0 ± 0.024 vs 0.1667 (deviation 4.51σ)
Duration fit vs published	0.046 d vs 0.095 d (deviation 1.22σ)
Mid-time O–C	5.8 min
Depth SNR	0.0
Residual scatter	2.89 %

Light curve

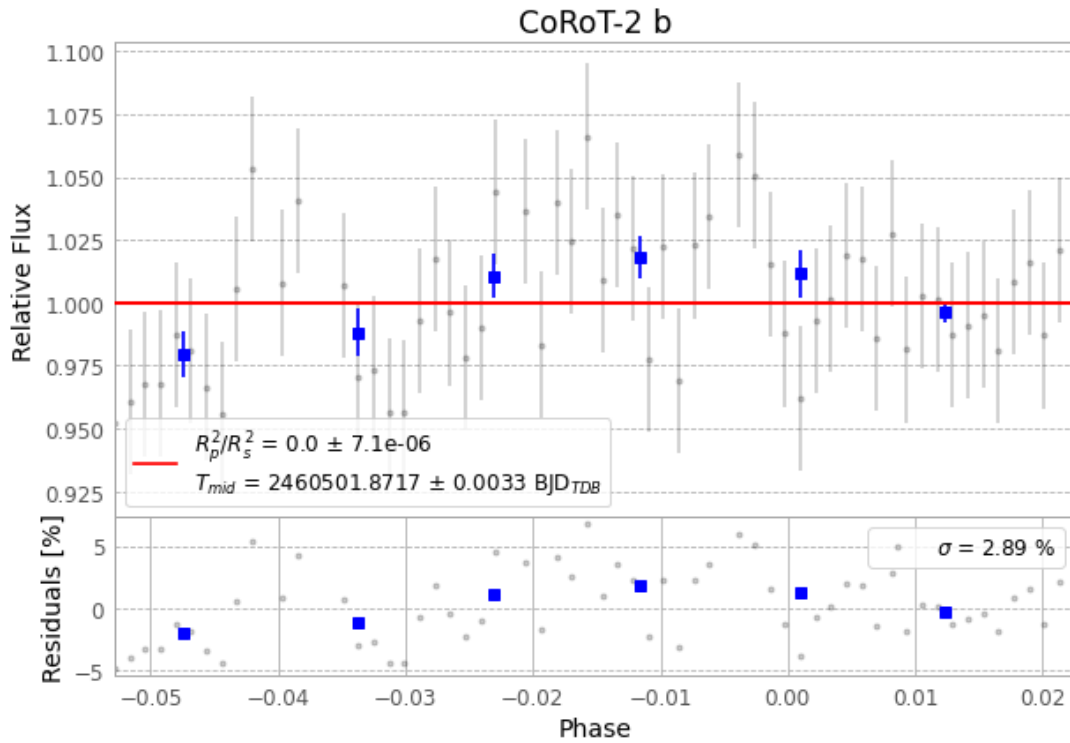


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-07-09.png (SHA-256 b10b8cd002a4571b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 332], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c3d20d4be4bb653...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

61 FITS frames (40 MB), CoRoT-2240710063315.FITS → CoRoT-2240710094215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240710.log (19ac9f267d34...), FinalLightCurve_CoRoT-2 b_2024-07-09.png (b10b8cd002a4...), FinalLightCurve_CoRoT-2 b_2024-07-09.csv (e950b7bd492f...)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:32Z.